ASSIGNMENT 8

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DOMAIN:DATA SCIENCE

```
#Creating Series from Array
import pandas as pd
import numpy as np
info = np.array(['P','a','n','d','a','s'])
a = pd.Series(info)
print(a)
```

```
0    P
1    a
2    n
3    d
4    a
5    s
dtype: object
```

```
#Create a Series from dict
#import the pandas library
import pandas as pd
import numpy as np
info = {'x' : 0., 'y' : 1., 'z' : 2.}
a = pd.Series(info)
```

```
print (a)
```

```
x    0.0
y    1.0
z    2.0
dtype: float64
```

```
#Create a Series using Scalar:
#import pandas library
import pandas as pd
import numpy as np
x = pd.Series('VIKAS', index=[0, 1, 2, 3])
print (x)
```

```
0    VIKAS
1    VIKAS
2    VIKAS
3    VIKAS
dtype: object
```

```
#Retrieving Index array and data array of a series object
import numpy as np
import pandas as pd
x=pd.Series(data=[2,4,6,8])
y=pd.Series(data=[11.2,18.6,22.5], index=['a','b','c'])
print(x.index)
```

```
print(x.values)
print(y.index)
print(y.values)
```

```
RangeIndex(start=0, stop=4, step=1)
[2 4 6 8]
Index(['a', 'b', 'c'], dtype='object')
[11.2 18.6 22.5]
```

```
#Retrieving Types (dtype) and Size of Type (itemsize)
import numpy as np
import pandas as pd
a=pd.Series(data=[1,2,3,4])
b=pd.Series(data=[4.9,8.2,5.6],
index=['x','y','z'])
print(a.dtype)
print(a.itemsize)
print(b.dtype)
print(b.itemsize)
```

```
int64
```

```
#Retrieving Shape
import numpy as np
import pandas as pd
a=pd.Series(data=[1,2,3,4])
```

```
b=pd.Series(data=[4.9,8.2,5.6],index=['x','y','z'])
print(a.shape)
print(b.shape)
```

```
(4,)
(3,)
```

```
#Retrieving Dimension, Size and Number of bytes:
import numpy as np
import pandas as pd
a=pd.Series(data=[1,2,3,4])
b=pd.Series(data=[4.9,8.2,5.6],
index=['x','y','z'])
print(a.ndim, b.ndim)
print(a.size, b.size)
print(a.nbytes, b.nbytes)
```

```
1 1
4 3
32 24
```

```
#Pandas Series.map()
import pandas as pd
import numpy as np
a = pd.Series(['Java', 'C', 'C++', np.nan])
print(a.map({'Java': 'Core'}) )
```

```
0    Core
1     NaN
2     NaN
3     NaN
dtype: object
```

```
import pandas as pd
import numpy as np
a = pd.Series(['Java', 'C', 'C++', np.nan])
a.map({'Java': 'Core'})
print(a.map('I like {}'.format, na_action='ignore')  )
```

```
0    I like Java
1       I like C
2     I like C++
3      NaN
dtype: object
```

P2

```
#Create a DataFrame from Dict of ndarrays/ Lists
# importing the pandas library
import pandas as pd
info = {'ID' :[101, 102, 103],'Department' :['B.Sc','B.Tech','M.Tech',]}
```

```
df = pd.DataFrame(info)

print (df)
```

```
 ID Department
0  101      B.Sc
1  102    B.Tech
2  103    M.Tech
```

```
#Create a DataFrame from Dict of Series:

# importing the pandas library

import pandas as pd

info = {'one' : pd.Series([1, 2, 3, 4, 5, 6], index=['a', 'b', 'c', 'd', 'e', 'f']),  'two' : pd.Series([1, 2, 3, 4, 5, 6, 7, 8], index=['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h'])}

d1 = pd.DataFrame(info)

print (d1)
```

```
one  two
a  1.0    1
b  2.0    2
c  3.0    3
d  4.0    4
e  5.0    5
f  6.0    6
g  NaN    7
h  NaN    8
```

```
# importing the pandas library
import pandas as pd
info = {'one' : pd.Series([1, 2, 3, 4, 5, 6], index=['a', 'b', 'c', 'd', 'e', 'f']),
   'two' : pd.Series([1, 2, 3, 4, 5, 6, 7, 8], index=['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h'])}
d1 = pd.DataFrame(info)
print (d1 ['one'])
```

```
a    1.0
b    2.0
c    3.0
d    4.0
e    5.0
f    6.0
g    NaN
h    NaN
Name: one, dtype: float64
```

```
# importing the pandas library
import pandas as pd
info = {'one' : pd.Series([1, 2, 3, 4, 5], index=['a', 'b', 'c', 'd', 'e']),
   'two' : pd.Series([1, 2, 3, 4, 5, 6], index=['a', 'b', 'c', 'd', 'e', 'f'])}
df = pd.DataFrame(info)
# Add a new column to an existing DataFrame object
print ("Add new column by passing series")
df['three']=pd.Series([20,40,60],index=['a','b','c'])
print (df)
```

```
print ("Add new column using existing DataFrame columns")
df['four']=df['one']+df['three']
print (df)
```

```
Add new column by passing series
   one  two  three
a  1.0    1   20.0
b  2.0    2   40.0
c  3.0    3   60.0
d  4.0    4    NaN
e  5.0    5    NaN
f  NaN    6    NaN
Add new column using existing DataFrame columns
   one  two  three  four
a  1.0    1   20.0  21.0
b  2.0    2   40.0  42.0
c  3.0    3   60.0  63.0
d  4.0    4    NaN   NaN
e  5.0    5    NaN   NaN
f  NaN    6    NaN   NaN
```

```
# importing the pandas library
import pandas as pd
info = {'one' : pd.Series([1, 2], index= ['a', 'b']),
   'two' : pd.Series([1, 2, 3], index=['a', 'b', 'c'])}
```

```
df = pd.DataFrame(info)
print ("The DataFrame:")
print (df)
# using del function
print ("Delete the first column:")
del df['one']
print (df)
# using pop function
print ("Delete the another column:")
df.pop('two')
print (df)
```

```
The DataFrame:
   one  two
a  1.0    1
b  2.0    2
c  NaN    3
Delete the first column:
   two
a    1
b    2
c    3
Delete the another column:
Empty DataFrame
Columns: []
Index: [a, b, c]
```

```
# importing the pandas library
import pandas as pd
info = {'one' : pd.Series([1, 2, 3, 4, 5], index=['a', 'b', 'c', 'd', 'e']),
  'two' : pd.Series([1, 2, 3, 4, 5, 6], index=['a', 'b', 'c', 'd', 'e', 'f'])}
df = pd.DataFrame(info)
print (df[2:5])
```

```
   one   two
c   3.0    3
d   4.0    4
e   5.0    5
```

```
# importing the pandas library
import pandas as pd
d = pd.DataFrame([[7, 8], [9, 10]], columns = ['x','y'])
d2 = pd.DataFrame([[11, 12], [13, 14]], columns = ['x','y'])
d = d.append(d2)
print (d)
```

```
     x     y
0    7     8
1    9    10
0   11    12
1   13    14
```

```
import pandas as pd
import numpy as np
```

```
info = pd.DataFrame({'col1':[7,1,8,3],'col2':[8,12,4,9]})

info_2 = info.sort_values(by='col2')

print(info_2)
```

	col1	col2
2	8	4
0	7	8
3	3	9
1	1	12

```
import pandas as pd

info_marks = pd.DataFrame({
    'name': ['Parker', 'Smith', 'William', 'Terry'],
    'Maths': [78, 84, 67, 72],
    'Science': [89, 92, 61, 77],
    'English': [72, 75, 64, 82]
})

# Use a valid existing path
output_path = r'C:\Users\HP\PycharmProjects\PythonProject\output.xlsx'

with pd.ExcelWriter(output_path) as writer:
    info_marks.to_excel(writer, index=False)

print('DataFrame is written successfully to the Excel File.')
```

```
'DataFrame is written successfully to the Excel File
```